

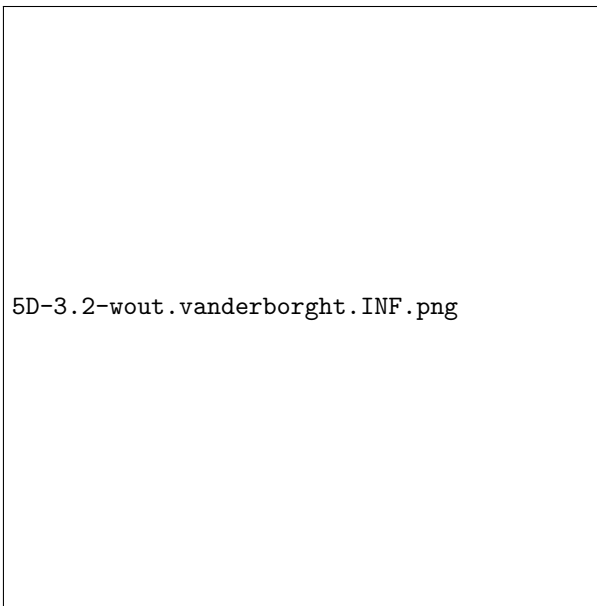
Auteur: Wout Vanderborght

Studierichting: Informatica

Oefeningenreeks 5D nr. 3.2

$$\begin{aligned} f(x) &= \frac{1}{x}, g(x) = \frac{1}{x^2} \text{ in } \left[\frac{1}{2}, 2 \right] \\ S &= \int_{\frac{1}{2}}^1 \left(\frac{1}{x^2} - \frac{1}{x} \right) + \int_1^2 \left(\frac{1}{x} - \frac{1}{x^2} \right) \\ &= \int_{\frac{1}{2}}^1 \frac{dx}{x^2} - \int_{\frac{1}{2}}^1 \frac{dx}{x} + \int_1^2 \frac{dx}{x} - \int_1^2 \frac{dx}{x^2} \\ &= \left[\frac{-1}{x} \right]_{\frac{1}{2}}^1 - [\ln |x|]_{\frac{1}{2}}^1 + [\ln |x|]_1^2 - \left[\frac{-1}{x} \right]_1^2 \\ &= 1 + \ln \frac{1}{2} + \ln 2 - \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

Tekening:



References